SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: Cryptography and Network Security
COURSE CODE: CSA5101

COURSE OUTCOME COVERED

CO1: Apply classical encryption techniques using symmetric and asymmetric ciphers to encrypt and decrypt data for the ensuring data security. (BL3, BL4)

Assessment Tool Weightage: 40%

QUESTIONS: 5 **Total Marks:50**

Annexure A

APPLICATION- BASED QUESTIONS Slot-A

Code of Conduct:

I Karanam Bhaskar Gnanesh (192324080) *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

K.B. Gnanesh

95/100

Annexure A

APPLICATION- BASED QUESTIONS

1. Scenario: Event Communication Security: A student organization is preparing to share confidential event details with its core members using a simple classical encryption method. During testing, a message "MEET AT NOON" is transformed into an encoded version using a fixed shift pattern.

a. Reconstruct how the encrypted message might appear if a shift of 4 positions is applied.
b. Explain how an authorized member, aware of the encoding rule, would retrieve the original message.

2. Scenario: Intercepted Suspicious Message: A cybersecurity intern intercepts the coded message: "GSRH RH Z HVXIVG" during network monitoring. The structure of the text suggests a simple substitution mechanism rather than complex encryption.

a. Deduce the most likely original message.
b. Justify your answer by analysing repeating patterns, letter structures, or known cipher characteristics.

3. Scenario: Secure Messaging System: A company deploys a keyword-based encryption technique for internal communication. An employee sends the word "HELLO" using the keyword "KEY" to ensure confidentiality.

a. Demonstrate how the encryption process transforms the message step-by-step.
b. Explain how the repeating keyword influences each character of the ciphertext.

4. Scenario: Suspicious Digital Asset Investigation: During a digital forensic audit, investigators suspect that a company logo file may contain hidden confidential information embedded within it. The file appears normal upon casual inspection.

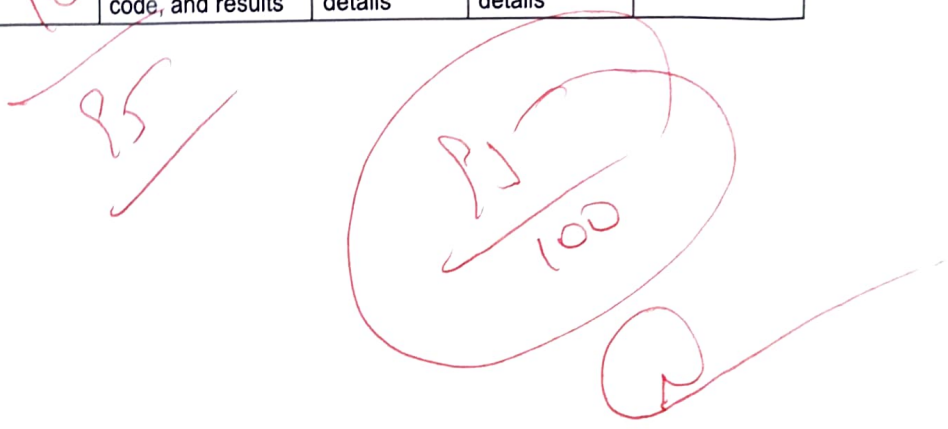
a. Analyze how investigators could detect the presence of hidden data within the image.
b. Recommend two techniques that could strengthen the security of such hidden communications against detection or extraction.

5. Scenario: Legacy Encryption in Corporate Communication: A legacy system in a company still uses a transposition-based encryption method with three levels of arrangement. A received message appears as: "WECRLTEERDSOEEFEAOCAIVDEN".

a. Determine the most probable original message by reversing the encryption process.
b. Critically evaluate why such a method is vulnerable in modern communication systems.
c. Propose one practical enhancement to improve the confidentiality of messages in this scenario.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25	Demonstrates thorough understanding and effectively integrates multiple concepts/technologies	Good understanding with proper integration of most concepts	Basic understanding; limited or partial integration	Poor understanding; unable to integrate concepts
Experimental Design	30	Well-planned, systematic implementation with optimal solution	Correct implementation with minor issues	Basic implementation with errors or inefficiencies	Incorrect or incomplete implementation
Use of Tools	20	Effective and advanced use of tools/technologies with proper configuration	Appropriate use of tools with correct execution	Limited or inconsistent use of tools	Incorrect or minimal use of tools
Analysis of Results	15	Insightful analysis with accurate interpretation and validation of results	Good analysis with reasonable interpretation	Basic analysis with limited insights	Poor or incorrect analysis of results
Documentation	10	Clear, well-structured documentation with diagrams, code, and results	Organized documentation with necessary details	Basic documentation with missing details	Poor or incomplete documentation



Karanam Bhaskar Gnanesh

192324080

B.Tech- AI&DS

~CSA5101

1. Caeser cipher

Caesar cipher is one of the simplest and oldest substitution encryption techniques. It was reportedly used by Julius Caeser to Protect military messages.

Advantages

1. Simple to understand - The encryption and decryption Process is very easy.
2. Easy to implement - It can be implemented with simple Programming logic.
3. fast encryption - It requires very little computation

Disadvantages

1. Very weak security - There are only 25 Possible shifts, So an attacker easily try all of them.
2. Vulnerable to brute-force attacks - Every possible key can be tested Quickly
3. Easy to break - Tools are even manual analysis can recover the plaintext quickly

1.

a. Caesar cipher

The Caesar cipher with a forward shift of 4

So apply key value is . 4

Caesar cipher :

Encryption : $E(P) = (P+K) \bmod (26)$

plain text : MEET AT NOON

cipher text : QIIX EX RSSR

1. b. Decryption : $D(C) = (C-K) \bmod (26)$

key = 4

cipher text : QIIX EX RSSR

plain text : MEET AT NOON

Scenario : Event Communication Security

A student organization wants to share confidential event details with its core members. To Provide basic community the organization uses a simple classical encryption technique based on fixed shift Pattern. The message given is:

plain text: MEET AT NOON

A shift of 4 Positions is applied to each alphabetic character.

M -> Q

E -> I

E -> I

T -> X

Therefore QIIX

A → €

T → X

∴ AT → €X

for final word

No → R

O → S

O → S

N → R

The completed encrypted message is :

@ II X € X RSS R

Encryption is

$C = (P + k) \bmod 26.$

Analyze how investigators could detect the Presence of hidden data within the image:

Investigators can examine the image using steganalysis first, they should inspect file metadata, filesize, format and unusual appended data. Next, they can compare pixel statistics, color-channel distributions and least-significant-bit (LSB) Patterns with those expected from a normal image. visual enhancement of individual color channels and bit planes can reveal abnormal Patterns. They can also check whether the image contains suspicious payloads or unexpected changes in pixel values. if available, a forensic steganalysis tools can automatic statistical tests and help identify the likely embedded method.

b. Recommend two techniques that could strengthe. the security of such hidden communications against detection or extraction.

Two techniques are recommended:

(1) Encrypt the secret data before embedding it, so that even if the hidden Payload is extracted it remains;

(2) use adaptive or randomized steganographic embedding with a secret key, distributing changes carefully across suitable image regions to reduce detectable statistical patterns.

Scenario : Suspicious Digital Asset Investigation

During a final digital forensic investigation, investigators suspect that a company logo contains hidden confédential information. The image appears completely normal when viewed casually, but it may contain secret info embedded it

This situation related to image steganography

steganography is the technique of hiding secret information inside another ordinary-looking medium such as image, audiofile, video or document.

a.) How investigate can detect data

Investigators can follow a systematic digital process to determine whether hidden information exists inside the image.

1. obtain the original image
2. Examine file properties
3. Analyze metadata
4. Analyze Pixel values.
5. Visualize Analysis
6. Histogram analysis.
7. Statistical analysis
8. Compare with a original Goal
9. use steganalysis tools.
10. Attempts data extraction.